Joint-Embedding Predictive Architectures and Fluid-Dynamic Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize point-wise error in raw observation space, where unforecastable jitter and scale disparities produce heavy-tailed residuals and frequent false alarms under automated calibration. Joint-Embedding Predictive Architectures (JEPAs) avoid this by forecasting compact latent summaries of future windows, yet they still collapse prediction errors into a single scalar score, allowing one or two drifting coordinates to inflate the threshold and suppress point-wise F1. We propose Laminar-JEPA, which pairs a multi-horizon JEPA with surgical weight-space quenching and a 224-bit circuit-breaker panel that replaces scalar scoring and tail-fitted thresholding. Each of 32 whitened latent coordinates is expanded via a Reynolds-inspired multi-horizon decomposition across four nested horizons into seven statistics (four horizon errors and their horizon mean, fluctuation, and endpoint difference), each monitored by an independently calibrated breaker with a nominal drift guard after removing mean predictor-bias drift and tempering high-variance precision directions on healthy calibration windows. On 19 development streams, Laminar-JEPA reaches 0.262 buffered point-wise F1, 66.4 percent buffered and 62.5 percent causal event recall, 1.6 percent causal false-alarm rate, and a 59.6-step detection delay; across look-ahead horizons from 1 to 64 steps, it doubles to quadruples scalar causal F1 down to zero reporting lag. Across 162 streams spanning 15 datasets, weight-space quenching cuts causal false alarms by 8.0 percent relative (median held-out false-alarm rate 1.0 percent and test false-alarm rate 1.6 percent), and on 118 untouched held-out streams a three-horizon panel raises causal point-wise F1 from 0.072 to 0.083 for scalar JEPAs to 0.143 and event recall from 51 to 73 percent. The panel requires roughly 1,000 calibration windows (300 effective independent samples), misses slow ramps, and needs a complementary channel-level branch for isolated single-sensor faults.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Fluid-dynamic decomposition, Boolean circuit-breaker panels, Representation learning.**

I. Introduction

Spacecraft, chemical plants, server farms, and wearable medical devices report health through dozens of continuous sensor channels [1]–[4]. When a valve sticks, a reaction wheel drifts, or a patient's gait freezes, an automated monitor must detect the fault promptly while learning healthy behavior from normal history alone, because labeled faults are rare. Equally important, the monitor must keep false alarms low during nominal operation so operators can trust its alerts.

Why Observation-Space Reconstructors Suffer High False-Alarm Rates: Deep detectors such as TimesNet [5], TranAD [6], and Anomaly Transformer [7] reconstruct the raw sensor signal and alert when reconstruction error is large. Raw telemetry, however, contains unforecastable sensor jitter, thermal drift, and disparate channel scales. Summed across 25 to 55 channels, these misses form a heavy-tailed residual with frequent spikes (Table I). Calibrated on healthy data via Streaming Peaks-Over-Threshold (SPOT) [8] and evaluated on later healthy data, reconstructors raise false alarms during 22%–31% of normal operation; a random alerter firing as often would catch 60.0% of anomaly events, exceeding TimesNet's 58.8%.

The Bottleneck of Scalar Score Aggregation: Joint-Embedding Predictive Architectures (JEPAs) [9], [10] avoid fitting raw noise by compressing each telemetry window into a 32-dimensional unit-sphere summary and predicting future summaries in latent space, cutting false alarms to 0.6%–2.4%. Yet existing latent detectors still sum all coordinate errors into one scalar score compared against one threshold. If one or two of the 32 coordinates shift slightly between training and calibration—variance-dominating coordinates we term *hijacker dials*—they dominate the sum and inflate the threshold, so faults on the remaining 30 coordinates go undetected: causal Point-F1 falls to 0.050–0.052 and average delay exceeds 240 steps.

*Four-Stage Architecture (*Laminar-JEPA*):* We replace scalar score aggregation with a four-stage architecture, *Laminar-JEPA* (Fig. 1). First, a causal dilated encoder maps context and target windows onto a 32-dimensional spherical latent space via layer-normalization (LayerNorm) [11] and forecasts four nested sub-horizons with Ledoit–Wolf Mahalanobis whitening. Second, a post-training *surgical weight-space quenching* pass on healthy calibration windows removes mean predictor-bias drift along the LayerNorm tangent space and tempers top precision-matrix eigendirections before scoring. Third, by structural analogy to Reynolds averaging in fluid mechanics [12] (without physical Navier–Stokes assumptions), we decompose each of the 32 coordinate error trajectories across the four nested horizons into seven kinematic views: four sub-horizon errors plus their horizon mean (drift), standard deviation

Manuscript received October 15, 2026. This work was supported by the Department of Electrical Engineering and Computer Science at Florida Atlantic University. No prior conference presentation or thesis material is included in this submission. *(Corresponding author: Andre Williams.)*

Andre Williams, Imadeldin Mahgoub, and Hamid Akbarian are with the Department of Electrical Engineering and Computer Science, Florida Atlantic University, Boca Raton, FL 33431 USA (e-mail: andrewilliam2018@fau.edu; mahgoubi@fau.edu; hakbaria@fau.edu).

Color versions of one or more figures in this article are available online at https://ieeexplore.ieee.org.

(fluctuation), and endpoint difference (acceleration). Fourth, these $7 \times 32 = 224$ statistics feed an independently calibrated *224-bit circuit-breaker panel* with per-dial calibration drift guards and 32 fast right-edge coincidence relays.

A. Contributions

We separate our evaluation into a 19/33-stream development cohort for ablations and a disjoint 118-stream untouched held-out cohort (within a 141/162-stream archive across 15 datasets) providing primary generalization evidence (Table II):

1) *Laminar-JEPA (224-Bit) across development and untouched held-out cohorts:* Replacing scalar thresholding with the 224-bit circuit-breaker panel and surgical weight-space quenching in *Laminar-JEPA* (Sections IV-C–IV-D) achieves the highest buffered Point-F1 (0.262), buffered event recall (66.4%), causal event recall (62.5%), fastest causal detection delay (59.6 steps vs. TimesNet's 62.2), and highest Dataset-Macro Point-F1 (0.139) on the 19/33 development streams while holding causal false-positive rate (FPR) at 1.6% (vs. 22.2%–24.3% for TranAD and TimesNet). Across 162 streams (15 datasets), surgical quenching in *Laminar-JEPA* cuts causal held-out FPR by 8.0% relative ($p = 0.0011$, dataset $p = 0.027$; median held-out FPR 1.03%, test FPR 1.59%), and on the 118 untouched held-out streams a three-horizon panel raises causal Point-F1 from 0.072–0.083 (1D JEPAs) to 0.143 ($p = 1.6\times10^{-13}$, dataset $p = 0.007$) and event recall from 51.2% to 72.8% (Tables III–VI).
2) *Mechanism isolation, quenching ablations, and multi-horizon latency scaling:* Matched-backbone ablations (272k–312k parameters, Table III) isolate LayerNorm kurtosis compression ($\gamma_2 : 6.10 \to 0.91$), learned latent targets (drift $D_{\mathrm{KS}} : 0.376 \to 0.214$), barycenter centering (regime-drift FPR share $71.3\% \to 18.7\%$), the 1-to-224 dial ladder, and bias recentering vs. precision tempering across $k_{\text{rogue}} \in \{0, \ldots, 12\}$. Retraining *Laminar-JEPA (224-Bit)* across target horizons $S \in \{1, 4, 16, 32, 64\}$ confirms $2.6\times$–$4.3\times$ higher causal Point-F1 and +33%–38% higher event recall than 1D scalar scoring down to $S = 1$ (zero lag) and $S = 4$ (3-step lag).
3) *False-alarm reduction and diagnostic fixed-budget recall:* On the 19 development streams, latent prediction holds autonomous causal FPR at 0.6%–1.6% versus 22.2%–24.3% for reconstructors ($N = 7$ dataset $p = 0.016$). Under a non-deployable post-hoc diagnostic control at $f^* = 1\%$ test FPR, the 224-bit panel catches 58.4% of anomaly events versus 21.7% for TimesNet and 14.9% for TranAD, and reaches 0.658 buffered precision-recall area under the curve (PR-AUC) on NASA contextual faults (Tables IV–V, Fig. 4).
4) *Calibration sample-size theory, single-channel branch, and null results:* Proposition 1 bounds the horizon-fluctuation concentration ratio $R_{\mathrm{LT}} \in [\frac{1}{M-1}, 1]$ (a structural representation property; scalar R_{LT} reweighting is null, $\Delta\mathrm{F1} = -0.005$). Auditing effective sample sizes $N_{\text{eff}} \in \{100, \ldots, 5{,}000\}$ with moving-block bootstrap confidence intervals establishes a minimum requirement of $N_{\text{eff}} \geq 300$ independent samples ($N_{\text{cal}} \geq 1{,}000$ windows) for empirical quantiles, while interquartile-range (IQR) and Beta-smoothed calibrators hold test FPR at 0.47%–1.32% when $N_{\text{eff}} \leq 250$. Across 17 multivariate streams and six single-sensor fault classes, a complementary channel-level branch raises single-channel event recall from 61.4%–80.7% to 93.0%–98.2%, whereas slow ramps remain undetected.

TABLE I
KEY TERMS.

Term	Meaning
Latent residual	Multi-horizon gap $\tilde{\mathbf{E}} \in \mathbb{R}^{4\times32}$ between predicted and observed 32D window summaries.
224-bit breaker	Boolean panel of 7×32 per-coordinate tripwires and 32 onset relays replacing 1D scalar thresholding.
Reynolds dials	Per-coordinate horizon mean $\lvert\bar{e}_c\rvert$ and fluctuation σ'_c across nested sub-horizons (32 dials each).
Right-edge relay	Coincidence switch $r_c(t)$ requiring terminal sub-block $[48, 64]$ and endpoint change or fluctuation to trip together.
Hijack guard	Nominal-only per-dial multiplier $\gamma_j^{\text{cal}}\gamma_j^{\text{sc}}$ damping variance-dominating coordinates (*hijacker dials*) on calibration data.
Surgical quenching	Post-training calibration pass removing mean predictor-bias drift and tempering top precision directions.
Buffered / Causal	Retrospective prefix-window voting vs. real-time trailing attribution at $\tau = t + S - 1$.
Calibration drift	Shift between calibration and later healthy residuals, measured by Kolmogorov–Smirnov distance D_{KS}.
Concentration ratio	Structural ratio $R_{\mathrm{LT}} \in [\frac{1}{M-1}, 1]$ measuring horizon-fluctuation energy in the leading eigenmode.

Roadmap. Section II reviews related work, Section III details the benchmark, Section IV formulates the model and panel, Section V defines the protocol, Sections VI–VII report results, and Section VIII concludes.

II. RELATED WORK

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly forecast sensor readings or reconstruct input windows in raw observation space, flagging timestamps where point-level error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive long short-term memory (LSTM) networks with non-parametric dynamic thresholding [4] model sequential dependencies directly on spacecraft channels. Recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [6] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [7] contrasts prior Gaussian kernel associations against series-level self-attention; and TimesNet [5] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods for 2D convolutional modeling. Because point-level losses penalize error across every channel equally, reconstructors fit stochastic sensor jitter and environmental fluctuations. In high-dimensional streams ($K \geq 25$), summing raw channel errors accumulates heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) residual spikes that trigger frequent false alarms under baseline shift [13].

Contrastive and Joint-Embedding Predictive Architectures: Self-supervised representation methods map windows into a compact latent space [14]. Contrastive detectors construct positive and negative pairs: temporal and contextual contrasting [15] and TS2Vec [16] maximize contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [10] contrasts full-window and context-only embeddings using injected synthetic anomalies; and DCdetector [17] employs dual patch-wise and in-patch attention without reconstruction. However, contrastive detectors depend on whether

synthetic perturbations or patch discrepancies match test-time fault morphology (0.088 and 0.097 stream Point-F1 for NCAD and DCdetector vs. 0.186 for latent prediction across 130 streams, Section VII-A). Joint-Embedding Predictive Architectures (JEPAs) [9], [18], [19] eliminate observation decoding and negative pairs by encoding context z_{ctx} and predicting the target embedding z_{tgt} from a separate target encoder, preventing collapse via exponential moving average (EMA) target updates [20] or variance-covariance penalties such as VICReg [21], Barlow Twins [22], and batch whitening [23].

Reynolds-Inspired Residual Decomposition and Circuit-Breaker Panels vs. Statistical Process Control: Multi-sensor telemetry reflects physical dynamical trajectories [24]. Neural ordinary differential equations (ODEs) [25], Physics-Informed Neural Networks (PINNs) [26], and Hamiltonian or Lagrangian networks [27], [28] enforce differential conservation laws on physical coordinates, but require known governing equations and suffer from stiff gradient pathologies on uncalibrated sensor arrays [29]. Koopman and operator-theoretic models [30]–[34] lift nonlinear dynamics into learned observable spaces. Rather than imposing partial differential equations on raw sensors, we apply a Reynolds-inspired multi-horizon residual decomposition [12], by analogy, to split multi-horizon latent residuals into per-coordinate horizon mean, fluctuation, and endpoint difference. Classical multivariate statistical process control (SPC) charts such as Hotelling's T^2 [35] and Multivariate Exponentially Weighted Moving Average (MEWMA) [36] quadratic-form-pool whitened coordinates into a single Mahalanobis scalar, just as deep detectors collapse residuals into a 1D score calibrated via Pickands–Balkema–de Haan extreme-value theory [37] and Generalized Pareto Distribution (GPD) fitting in SPOT [8], where a single coordinate with out-of-sample variance inflation under serial correlation [38] dominates the scalar threshold. Conversely, banks of univariate Shewhart or Cumulative Sum (CUSUM) charts [39] monitor raw channels or single statistics under Bonferroni or false-discovery-rate (FDR) multiplicity corrections [40], but lack multi-horizon kinematic perspectives, per-dial drift quarantine guards, and right-edge coincidence relays; soft attention ensemble voting across dials adds no benefit over Boolean quorums (Table III(b)). We evaluate all detectors using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm curves [13], [41], [42].

III. Multi-Sensor Internet-of-Things and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [43], [44], TimeEval [13], and mTSBench [45], expanded in archive-wide evaluations to 141 qualifying multivariate streams across 12 datasets and 162 streams across 15 datasets (Table II). Thirty-three streams enter the primary protocol (19 non-GHL development streams and 14 GHL streams); 12 are reserved with explicit accounting below. The suite spans four domains: *Spacecraft and solar telemetry* covers six SMAP streams (K=25 channels), six MSL rover streams (K=55, four primary and two short-nominal), and SWAN-SF solar magnetograms [46] (K=38), where NASA's archive [4] classifies each SMAP/MSL sequence as an out-of-range *point* fault or in-range *contextual* fault (Table V(a)). *Wearable kinematics* covers six Daphnet Freezing-of-Gait (FoG) streams [43] at 64 Hz (K=9, 72 episodes, where S=64 equals 1 s). *Industrial and infrastructure telemetry* covers 14 Gasoil Heating Loop (GHL) streams (K=16, 26 faults), the Genesis pick-and-place testbed (K=18), GECCO water chemistry at 1-minute resolution (K=9, 37 anomalies), Metro I-94 hourly traffic (K=5), Room-Occupancy sensing [47] (K=5), CalIt2 building flow [44] (K=2), and six CICIDS network streams (K=72). Section VI-B evaluates six single-channel fault classes across all 17 multivariate development streams ($K > 1$).

Each stream provides nominal training sequences ($N_{\text{tr}} \in [2{,}074, 120{,}000]$) and labeled test sequences ($N_{\text{test}} \in [2{,}158, 114{,}001]$). Channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2}. Each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by a 64-step isolation gap (Section V-A). Primary 33-stream inclusion required $N_{\text{cal}} \geq 500$ ($T_{\text{nom}} \geq 3{,}769$) and $K \leq 55$, $N_{\text{test}} \leq 1.2 \times 10^5$ (Table IV(c)); among the 12 reserved streams, five have $T_{\text{nom}} \in [764, 1{,}492]$ ($N_{\text{cal}} \leq 101$ across CalIt2, MSL C-2, MSL D-16, and two Room-Occupancy streams), six CICIDS streams have K=72, $N_{\text{test}} \geq 1.36 \times 10^5$, and GECCO ($K$=9) enters the archive cohorts. To eliminate subset-selection bias, Table VI evaluates all 141 qualifying multivariate streams across 12 datasets ($N_{\text{cal}} \geq 100$ and ≥ 1 test anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, 13 OPPORTUNITY, GECCO, Genesis, Pooled Server Metrics (PSM), CreditCard, Metro, and SWAN) at three seeds, of which 118 streams were never inspected during development, and evaluates four-horizon Laminar-JEPA across 162 streams spanning 15 datasets (adding 14 GHL, Secure Water Treatment (SWaT), Water Distribution (WADI), and reserve streams; 139 untouched streams).

Two caveats bound external validity: (i) SMAP and MSL contain documented labeling flaws [42]; and (ii) streams within multi-stream archives share hardware platforms. We therefore treat dataset clusters ($N \in \{7, 9, 12, 15\}$) as co-primary statistical units alongside streams, reporting Dataset-Macro means, cluster-bootstrap confidence intervals (CI), exact dataset sign-flip permutation tests, and leave-one-dataset-out (LODO) verification (Section V-A).

IV. Methodology: Joint-Embedding Predictive Architectures and 224-Bit Circuit-Breaker Panels

A. Stage 1: Multiscale Windowing, Multi-Depth TCN Hierarchy Tapping, and Barycenter Centering

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T \times K}$ recording K sensor channels over T timesteps. Fig. 1 illustrates the four-stage pipeline from raw telemetry to binary alarm decisions. In Stage 1, a sliding window of length $W = C + S = 320$ advances at stride 1 and partitions the stream at

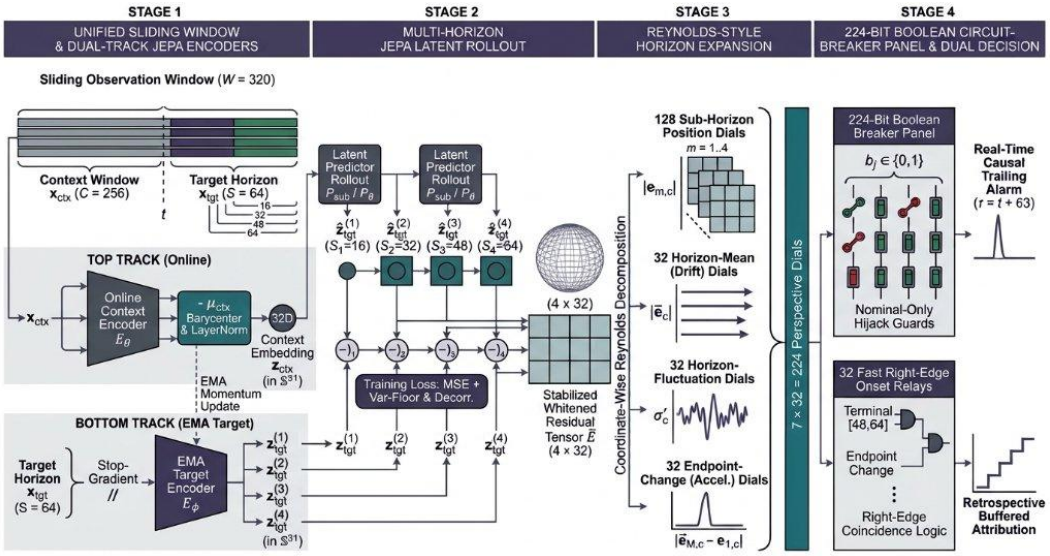


Fig. 1. Overview of the four-stage Multi-Horizon Reynolds JEPA and 224-bit circuit-breaker pipeline: (1) multiscale dyadic windowing, 3-depth TCN hierarchy tapping, and barycenter centering; (2) nested 4-horizon latent rollout under a variance-floor and decorrelation objective; (3) calibration-stabilized Mahalanobis whitening, optional surgical weight-space quenching, and Reynolds-inspired expansion into $7 \times 32 = 224$ dials; and (4) 224-bit Boolean circuit-breaker tripwires and 32 right-edge onset relays.

index t into a *context window* $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C \times K}$ ($C = 256$) and a *suspect target horizon* $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t + S, :] \in \mathbb{R}^{S \times K}$ ($S = 64$, up to decision timestamp $\tau = t + S - 1$). For $K > 1$, a causal depthwise 1D convolutional stem filters each channel prior to 1×1 mixing ($K \leq 1$ uses an identity bypass). Channels are decomposed across $J = 4$ smooth dyadic shells on a 256-point real fast Fourier transform (rFFT) grid normalized into an exact amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ within 1.2×10^{-7}), processed by per-shell causal 1D convolutions (kernels $\{15, 9, 5, 3\}$), and recombined via a softmax router into a causal dilated temporal convolutional network (TCN) [48].

Unlike reconstructive detectors that minimize pointwise observation error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ and inherit heavy-tailed sensor noise (Sections I–II), we map windows into a compact 32-dimensional latent space ($D = 32$) using a Joint-Embedding Predictive Architecture (JEPA) [9]. Dual online (E_θ) and target (E_ϕ) encoders map $\mathbf{x}_{\text{ctx}}$ and $\mathbf{x}_{\text{tgt}}$ onto the sphere $\sqrt{D}\mathbb{S}^{D-1}$, where target weights ϕ track online weights θ via an exponential moving average (EMA), $\phi_k \leftarrow m\phi_{k-1} + (1 - m)\theta_k$ with fixed momentum $m = 0.996$ (0.995 in Cohorts 3–4) [18], [20]. Because deep dilated TCN blocks smooth out sharp onset spikes while non-stationary streams shift the latent mean, Multi-Horizon Reynolds JEPA introduces two structural operations inside both encoders:

1) *Multi-depth functional hierarchy tapping:* A tap head extracts $L_{\text{tap}} = 3$ intermediate feature maps $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B \times T_w \times d_{\text{model}}}$ ($d_{\text{model}} = 48$) from the shallow dyadic front-end, the mid-depth TCN block, and the terminal TCN block. For each tap ℓ, the channel-centered terminal state and temporal root-mean-square (RMS) energy are concatenated into $\mathbf{h}_{\text{tap}} \in \mathbb{R}^{B \times 6d_{\text{model}}}$, projected into $\mathbb{R}^D$, and fused with the deep representation $\mathbf{z}_{\text{deep}}$ via a bounded gate $g = 0.5\,\sigma(\theta_g) \leq 0.5$:

$$\mathbf{z}_{\text{fused}} = (1 - g)\,\mathbf{z}_{\text{deep}} + g\ \text{LN}\big(W_2\,\phi_{\text{GELU}}(\text{LN}(W_1\mathbf{h}_{\text{tap}}))\big). \tag{1}$$

2) *Synchronized barycenter centering:* An online running centroid $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ updated at rate 0.05 on training batches is copied synchronously to the EMA target tap head and subtracted from both context and target fused vectors before terminal LayerNorm projection $z = \text{LN}(\mathbf{z}_{\text{fused}} - \bar{\boldsymbol{\mu}}_{\text{ctx}}) \in \sqrt{D}\mathbb{S}^{D-1}$, canceling common-mode baseline drift between windows.

B. Stage 2: Nested Causal Multi-Horizon Rollout and Variance-Floor Decorrelation Objective

In the single-horizon baseline (*TS-JEPA*), a two-layer multilayer perceptron (MLP) predictor $P_\theta : \mathbb{R}^D \to \mathbb{R}^D$ maps z_{ctx} to a single full-window forecast $\hat{z}_{\text{tgt}} = P_\theta(z_{\text{ctx}})$ trained with predictive MSE plus a log-determinant volume term $-\frac{\beta}{D}\ln\det(\boldsymbol{\Sigma} + \epsilon\mathbf{I})$ on predicted and context batches ($\beta = 0.05$), a smooth relative of VICReg variance-covariance penalties [21]. Terminal LayerNorm bounds representations on $\sqrt{D}\mathbb{S}^{D-1}$ and compresses residual kurtosis, but pooling the $S = 64$ target window into one vector averages out short-lived sub-horizon transients.

To resolve sub-window dynamics across target horizon S (default $S = 64$, or $S \in \{1, 4, 16, 32, 64\}$ in Section VII-C), Stage 2 of *Multi-Horizon Reynolds JEPA* partitions $\mathbf{x}_{\text{tgt}}$ into $M = 4$ nested causal prefixes $\mathbf{x}_{\text{tgt}}^{(m)} = \mathbf{x}_{\text{tgt}}[0 : S_m, :]$ of lengths $S_m = \max(1, \lfloor mS/M \rfloor)$ ($\{16, 32, 48, 64\}$ at $S = 64$). The EMA target encoder E_ϕ encodes each prefix into $z_{\text{tgt}}^{(m)} \in \mathbb{R}^D$. Starting from $\hat{z}_{\text{tgt}}^{(0)} \equiv z_{\text{ctx}}$, a sub-horizon predictor P_{sub} rolls out sequential forecasts $\hat{z}_{\text{tgt}}^{(m)} = \text{LN}\big(P_{\text{sub}}(\hat{z}_{\text{tgt}}^{(m-1)} + \mathbf{p}_m)\big)$ for $m \in \{1, \ldots, M - 1\}$ using learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, while the full-horizon anchor P_θ forecasts $\hat{z}_{\text{tgt}}^{(M)} = \text{LN}\big(P_\theta(z_{\text{ctx}} + \mathbf{p}_M)\big)$. With horizon weights $w_m = \frac{1}{2(M-1)}$ ($m < M$) and $w_M = \frac{1}{2}$, training minimizes multi-horizon mean-squared error (MSE) plus a variance-floor and decorrelation penalty:

$$\mathcal{L}_{\text{MHR}} = \sum_{m=1}^{M} \frac{w_m}{D}\Big(\big\|\hat{z}_{\text{tgt}}^{(m)} - \text{sg}\big(z_{\text{tgt}}^{(m)}\big)\big\|_2^2 + \big\|\boldsymbol{\Delta}_\gamma\big(\hat{z}_{\text{tgt}}^{(m)}\big)\big\|_F^2\Big), \tag{2}$$

where $\mathbf{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ penalizes low variance and cross-covariance of centered batch predictions $\mathbf{Z}_c = \mathbf{Z} - \mathbf{1}_B\bar{\mathbf{z}}^\top$ with empirical covariance $\mathbf{\Sigma}(\mathbf{Z}) = (B-1)^{-1}\mathbf{Z}_c^\top\mathbf{Z}_c$ and marginal standard deviations $\sigma_d(\mathbf{Z}) = \sqrt{\Sigma_{dd}(\mathbf{Z}) + \epsilon}$ ($\gamma = 0.75$, $\epsilon = 10^{-4}$), defined by $[\mathbf{\Delta}_\gamma(\mathbf{Z})]_{dd} = \sqrt{\max(0, \gamma - \sigma_d(\mathbf{Z}))}$ and $[\mathbf{\Delta}_\gamma(\mathbf{Z})]_{dj} = \frac{1}{\sqrt{2}}\Sigma_{dj}(\mathbf{Z})$ ($d \neq j$). This enforces a variance floor plus decorrelation (full whitening occurs in Stage 3 via Ledoit–Wolf shrinkage), remaining well-conditioned when $B = 16 < D = 32$ and raising effective representation rank $\mathrm{RankMe}(z_{\text{tgt}})$ from 11.6 to 19.8 [49]. Archive-wide evaluations test three-horizon *Speculative-JEPA* across 141 streams (prefixes $\{16, 32, 64\}$, weights $(0.20, 0.25, 0.55)$, and the 48-step residual interpolated from 32 and 64, giving 192 distinct statistics) and four-horizon *Laminar-JEPA* across 162 streams (15 datasets).

C. Stage 3: Calibration-Stabilized Whitening, Surgical Weight-Space Quenching, and 224-Dial Expansion

Why does collapsing a multi-horizon residual tensor into a single 1D score degrade point-wise F1? Let $\mathbf{E}(t) = [\mathbf{e}_1(t), \ldots, \mathbf{e}_M(t)]^\top \in \mathbb{R}^{M\times D}$ stack the $M = 4$ raw sub-horizon prediction residuals $\mathbf{e}_m(t) = z_{\text{tgt}}^{(m)}(t) - \hat{z}_{\text{tgt}}^{(m)}(t)$ across all $MD = 128$ coordinates. Fitting the Ledoit–Wolf shrinkage precision matrix $\mathbf{\Omega} = (\hat{\mathbf{\Sigma}}_{\text{res}} + 10^{-3}\mathbf{I}_{MD})^{-1/2} \in \mathcal{S}_{++}^{MD}$ [50] and mean $\bar{\boldsymbol{\mu}}_{\text{res}} \in \mathbb{R}^{MD}$ on nominal training windows produces Mahalanobis-whitened residuals $\mathbf{E}^{\text{white}}(t) \in \mathbb{R}^{M\times D}$ with unit training variance per coordinate. On unseen nominal calibration windows, however, one or two whitened directions often experience out-of-sample variance growth (*hijacker dials*) where $\mathbf{\Omega}$ amplifies small covariance shifts. Summing all 128 coordinates into a 1D scalar score allows those one or two drifting coordinates to dominate the calibration tail and inflate the scalar threshold. Stage 3 prevents both precision amplification and 1D collapse through two steps:

1) *Surgical weight-space quenching (*Laminar-JEPA*) and calibration-stabilized whitening:* Inverting $\mathbf{\Omega}$ and predictor Jacobians without data reconstructs the 224-dial coordinate roles at 88.4% accuracy (162 streams) and shows that stochastic gradient descent traps 12.7% of horizon-embedding energy in the saw-tooth mode $\boldsymbol{\phi}_3 = [-1, +3, -3, +1]^\top/\sqrt{20}$. A constant baseline shift between training and calibration offsets the mean residual $\bar{\mathbf{e}}_m = \mathbb{E}_{\text{cal}}[\mathbf{e}_m(t)]$ on $\sqrt{D}\mathbb{S}^{D-1}$, while covariance rotation inflates variance along steep precision eigenvectors ($\lambda_k \gg 1$). Because terminal LayerNorm maps pre-normalized outputs $\mathbf{u}_m$ onto $\sqrt{D}\mathbb{S}^{D-1}$ with first-order Jacobian $\mathbf{J}_{\text{LN}}(\hat{\boldsymbol{\mu}}_m) = (\sqrt{D}/\sigma_{\text{pre},m})(\mathbf{I}_D - \hat{\boldsymbol{\mu}}_m\hat{\boldsymbol{\mu}}_m^\top)$, shifting each predictor's terminal bias by $\Delta\mathbf{b}_m = (\sigma_{\text{pre},m}/\sqrt{D})(\mathbf{I}_D - \hat{\boldsymbol{\mu}}_m\hat{\boldsymbol{\mu}}_m^\top)\bar{\mathbf{e}}_m$ cancels the stationary tangent-space calibration drift to first order while leaving time-varying anomaly excursions $\mathbf{e}_m(t) - \bar{\mathbf{e}}_m$ invariant. Simultaneously, tempering $\mathbf{\Omega}$ via smooth saturation $\tilde{\lambda}_k = \lambda_k/(1+\lambda_k/\lambda_{\max})$, damping output rows on the top $k_{\text{rogue}} \leq 4$ variance-inflated coordinates, and decoupling their off-diagonal precision entries prevent an unstable direction from contaminating clean coordinates (accepted only when late-half calibration trips do not increase; ablations in Section VI-A). Any coordinate with $\mathrm{RMS}_{\text{cal}}^{\text{white}}(m,c) > 1.10$ is then blended toward $\hat{e}_{m,c}^{\text{raw}}(t) = e_{m,c}(t)/(\mathrm{RMS}_{\text{cal}}^{\text{raw}}(m,c) + 10^{-4})$:
$$\tilde{e}_{m,c}(t) = w_{m,c}\, e_{m,c}^{\text{white}}(t) + (1 - w_{m,c})\, \hat{e}_{m,c}^{\text{raw}}(t), \quad w_{m,c} = \frac{1}{1 + 2\max\big(0, \mathrm{RMS}_{\text{cal}}^{\text{white}}(m,c) - 1.10\big)^2}. \tag{3}$$
2) *Reynolds-inspired multi-horizon residual decomposition into $7D = 224$ dials:* By structural analogy to Reynolds decomposition in fluid turbulence [12] (applied here along the discrete lookahead axis $m \in \{1, \ldots, M\}$ rather than a continuous fluid field), we expand $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{4\times 32}$ into $7D = 224$ non-negative dials $\mathbf{d}(t) \in \mathbb{R}^{224}_{\geq 0}$: (a) *sub-horizon position magnitudes ($4D = 128$ dials)* $d_{\text{pos},m,c}(t) = |\tilde{e}_{m,c}(t)|$ ($m \in \{1..4\}$); (b) *horizon-mean (drift) dials ($D = 32$)* $d_{\text{drift},c}(t) = |\bar{e}_c(t)| = |\frac{1}{M}\sum_{m=1}^M \tilde{e}_{m,c}(t)|$; (c) *horizon-fluctuation dials ($D = 32$)* $d_{\text{fluc},c}(t) = \sigma'_c(t) = \sqrt{\frac{1}{M}\sum_{m=1}^M(\tilde{e}_{m,c}(t) - \bar{e}_c(t))^2}$; and (d) *endpoint-change (acceleration) dials ($D = 32$)* $d_{\text{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$.

Proposition 1 characterizes this multi-horizon residual representation, showing that the drift and fluctuation dials orthogonally partition sub-horizon energy and that the horizon-fluctuation spectrum lies on a bounded interval (a structural representation property rather than a guarantee of detection accuracy).

Proposition 1 (Structural Characterization of Multi-Horizon Residual Energy and Spectral Concentration)**.** *Let $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ ($M \geq 2$) denote the stabilized multi-horizon residual field with horizon mean $\bar{\mathbf{e}} = M^{-1}\tilde{\mathbf{E}}^\top\mathbf{1}_M \in \mathbb{R}^D$ and zero-mean fluctuation matrix $\mathbf{E}' = \tilde{\mathbf{E}} - \mathbf{1}_M\bar{\mathbf{e}}^\top \in \mathbb{R}^{M\times D}$. Then: (i)* Coordinate-wise energy orthogonality and endpoint bound: *For every coordinate $c \in \{1, \ldots, D\}$, $\frac{1}{M}\sum_{m=1}^M d_{pos,m,c}^2(t) = d_{drift,c}^2(t) + d_{fluc,c}^2(t)$, and $d_{accel,c}(t) \leq \sqrt{2M}\, d_{fluc,c}(t)$. (ii)* Spectral concentration bounds for the horizon Gram matrix: *Let $\mathbf{G} = \mathbf{E}'\mathbf{E}'^\top \in \mathcal{S}_+^M$ have ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$ ($\mathbf{E}'^\top\mathbf{1}_M = \mathbf{0}$). For any $\gamma > -1/2$, the spectral concentration ratio (following Lieb–Thirring power-sum moments [51]),*
$$R_{LT}(\mathbf{E}') = \frac{\lambda_M^{\gamma+1/2}}{\sum_{m=1}^M \lambda_m^{\gamma+1/2}} \in \Big[\frac{1}{M-1}, 1\Big] \quad (\mathbf{E}' \neq \mathbf{0}), \tag{4}$$
is scale-invariant, attaining $\frac{1}{M-1}$ ($= \frac{1}{3}$ for $M = 4$) iff fluctuation energy is isotropic across $M - 1$ modes, and 1 iff $\mathrm{rank}(\mathbf{E}') = 1$. Empirically, scalar reweighting by R_{LT} yields a null effect (Section VI).

Proof. Expanding $\frac{1}{M}\sum_{m=1}^M(\bar{e}_c + e'_{m,c})^2$ with $\sum_{m=1}^M e'_{m,c} = 0$ yields the energy identity, and Cauchy–Schwarz on $e'_{M,c} - e'_{1,c}$ gives $d_{\text{accel},c}^2 \leq 2(e'^2_{M,c} + e'^2_{1,c}) \leq 2M\, d_{\text{fluc},c}^2$, proving (i). For (ii), $\mathbf{G}\mathbf{1}_M = \mathbf{0}$ forces $\lambda_1 = 0$, so $\lambda_M^{\gamma+1/2} \leq \sum_{m=2}^M \lambda_m^{\gamma+1/2} \leq (M-1)\lambda_M^{\gamma+1/2}$, yielding (4). □

D. Stage 4: Zero-Collapse 224-Bit Circuit-Breaker Panel, Onset Relays, and Temporal Attribution

In Stage 4 (Fig. 1), each of the 224 dials receives its own binary circuit breaker calibrated on nominal data. Because all 224 dials derive from the same 4×32 residual field, they are

correlated by construction (median effective rank 65 of 224 on the 19 development streams; simultaneous trip counts have $7.2\times$ the variance of an independence model), so the quorums below are calibrated directly on joint nominal trip counts:

1) *Per-dial standardization and nominal hijack guards:* Each dial $j \in \{1,\dots,224\}$ (and $j \in \{1,\dots,128\}$ in the 128-bit sub-block panel) is standardized by its nominal calibration median m_j and scaled median absolute deviation (MAD) $s_j = 1.4826\,\mathrm{MAD}(d_{j,\mathrm{cal}}) + 10^{-4}$ into $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. To quarantine any drifting dial individually, each tripwire τ_j scales its 98.5th calibration percentile by a two-factor nominal hijack guard $\gamma_j = \gamma_j^{\mathrm{cal}}\,\gamma_j^{\mathrm{sc}}$:
$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\mathrm{cal}})\,\gamma_j^{\mathrm{cal}}\,\gamma_j^{\mathrm{sc}}, \quad (5)$$
where γ_j^{cal} is the ratio of second-half to first-half calibration 98th percentiles whenever > 1.35 (1 otherwise), and γ_j^{sc} is $(m_j + s_j)/(1.5\,s_{\mathrm{group},j})$ whenever > 1.25 (1 otherwise, where $s_{\mathrm{group},j}$ is the perspective-group median of $m_j + s_j$).
2) *32 fast right-edge onset relays:* Each coordinate $c \in \{1,\dots,32\}$ defines a stricter edge switch $b^{\mathrm{edge}}_{p,c}(t) = \mathbb{I}\big(z_{p,c}(t) > q_{0.993}(z_{p,c,\mathrm{cal}})\,\gamma_{p,c}^{1.25}\big)$ and fires its onset relay $r_c(t) \in \{0,1\}$ when the terminal sub-block ($m = 4$) coincides with endpoint acceleration, fluctuation, or the third sub-block ($m = 3$):
$$r_c(t) = b^{\mathrm{edge}}_{\mathrm{pos},4,c}(t) \wedge \big(b^{\mathrm{edge}}_{\mathrm{accel},c}(t) \vee b^{\mathrm{edge}}_{\mathrm{fluc},c}(t) \vee b^{\mathrm{edge}}_{\mathrm{pos},3,c}(t)\big). \quad (6)$$
3) *Causal trailing vs. prefix-routed buffered attribution and Boolean quorums:* Under *causal trailing attribution*, window switches at index t are emitted solely at the right endpoint $\tau = t + S - 1$ ($\mathcal{F}_\tau$-measurable, with structural reporting lag $S-1$ vs. empirical detection delay $\Delta t = t_{\mathrm{alm}} - t^*$, Fig. 2). Under *retrospective buffered attribution*, each perspective votes only over the steps inside $[t, t+S)$ it physically observes ($[t, t+S_m)$ for sub-block m, $[t, t+S)$ for drift and fluctuation, and $[t+S_3, t+S_4)$ for acceleration). Summing active switches at τ gives counts $N_{224}(\tau)$, $N_{128}(\tau)$, and $N_{\mathrm{edge}}(\tau)$, with combined count $N_{\mathrm{tot}}(\tau) = N_{224}(\tau) + N_{128}(\tau) + 2N_{\mathrm{edge}}(\tau)$ for PR-AUC ranking. Binary alarms trigger when any counter exceeds its calibration quorum:
$$\hat{y}(\tau) = \mathbb{I}\big(N_{224}(\tau) > K_{224} \;\vee\; N_{128}(\tau) > K_{128} \;\vee\; N_{\mathrm{edge}}(\tau) > K_{\mathrm{edge}}\big), \quad (7)$$
where $K_{\mathrm{edge}} = \max\big(1.25\,q_{0.999}(N_{\mathrm{edge,cal}}),\,2.5\big)$, *Laminar-JEPA (224-Bit)* sets $K_{224} = \max\big(1.12\,q_{0.995}(N_{224,\mathrm{cal}}),\,5.0\big)$ and $K_{128} = \max\big(1.12\,q_{0.995}(N_{128,\mathrm{cal}}),\,3.5\big)$, and *224-Bit Strict* sets $K_{224} = \max\big(1.18\,q_{0.9985}(N_{224,\mathrm{cal}}),\,5.5\big)$ and $K_{128} = \max\big(1.20\,q_{0.9985}(N_{128,\mathrm{cal}}),\,4.0\big)$.

1D Scalar Collapse Controls: To compare against 1D scalar collapse on identical backbones (Section VI), we evaluate the saliency gate $g_d(\mathbf{v}) = (1-\alpha)\|\mathbf{v}\|_2 + \alpha\sqrt{d\sum_{i=1}^{d} w_i v_i^2}$ with $w_i \propto \exp(|v_i|/(\tau_g \sigma_{\mathbf{v}}))$ ($\sigma_{\mathbf{v}} = \|\mathbf{v}\|_2/\sqrt{d} + 10^{-6}$, $\alpha = \tau_g = 0.5$) . Single-horizon TS-JEPA scores $\mathcal{S}_{\mathrm{TS}}(t) = g_D(\mathbf{e}(t))$, and 1D Multi-Horizon Reynolds JEPA scores $\mathcal{S}_{\mathrm{env}}(t) = \sqrt{\frac{1}{2}(g_{MD}(\tilde{\mathbf{e}}(t))^2 + \max_m g_D(\tilde{\mathbf{e}}_m(t))^2)}$ modulated by $\mathcal{S}_{\mathrm{LT}}(t) = \mathcal{S}_{\mathrm{env}}(t)\big(1+\alpha_{\mathrm{LT}}\max(0,\tanh(0.5\,z_{\mathrm{LT}}(t)))\big)$ ($\alpha_{\mathrm{LT}} \in \{0.12, 0\}$), calibrated via SPOT [8], [37], [52] ($u = p_{98}, q = 10^{-3}$).

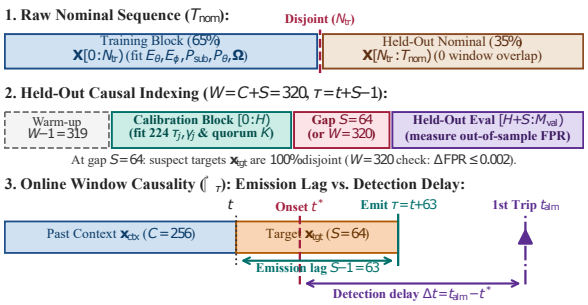


Fig. 2. Two-stage held-out evaluation protocol ($C{=}256, S{=}64, W{=}320$): (Row 1) chronological train/validation split, (Row 2) boundary-isolated nominal calibration and evaluation blocks fitting the 224 dial tripwires τ_j, hijack guards γ_j, and quorums ($K_{224}, K_{128}, K_{\mathrm{edge}}$), and (Row 3) structural emission lag ($S{-}1$) versus empirical fault-detection delay (Δt).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup, Two-Stage Nominal Calibration, and Statistical Design

All architectures operate on a streaming pipeline with context window $C = 256$ and target horizon $S = 64$ ($W = C + S = 320$, or $S \in \{1, 4, 16, 32, 64\}$ in Section VII-C), trained on nominal windows at stride 16 and evaluated at stride 1. Context and target windows are encoded into $D = 32$ spherical representations by the multiscale front-end and causal dilated TCN, fused with the three-depth tap head (1), and centered by the running context barycenter $\bar{\boldsymbol{\mu}}_{\mathrm{ctx}}$ prior to terminal LayerNorm. Models are trained using AdamW [53] (learning rate 10^{-3}, weight decay 10^{-4}) with zero per-dataset tuning: batch 16 for 15–20 epochs on the 19- and 33-stream cohorts, and batch 16–64 for 10–12 epochs with gradient clipping at 1.0 and cosine annealing on the archive cohorts.

a) Two-Stage Nominal Calibration Protocol and Boundary Isolation: Each stream's nominal sequence $\mathbf{X}_{\mathrm{nom}} \in \mathbb{R}^{T_{\mathrm{nom}} \times K}$ is partitioned chronologically (Fig. 2, Row 1) into a *training block* $\mathbf{X}_{\mathrm{tr}} = \mathbf{X}_{\mathrm{nom}}[0 : N_{\mathrm{tr}})$ ($N_{\mathrm{tr}} = \lfloor 0.65\,T_{\mathrm{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\mathrm{val}} = \mathbf{X}_{\mathrm{nom}}[N_{\mathrm{tr}} : T_{\mathrm{nom}})$ ($N_{\mathrm{val}} = T_{\mathrm{nom}} - N_{\mathrm{tr}}$): (1) *Stage A (Training fit on $\mathbf{X}_{tr}$):* Standardization parameters, encoder/predictor weights, running barycenter $\bar{\boldsymbol{\mu}}_{\mathrm{ctx}}$, and precision matrix $\boldsymbol{\Omega} \in \mathcal{S}_{++}^{MD}$ are fit strictly on $\mathbf{X}_{\mathrm{tr}}$ (in archive runs, early stopping uses unsupervised predictive loss on the calibration block without labels). (2) *Stage B (Circuit-breaker calibration and held-out verification on $\mathbf{X}_{val}$):* Sliding windows of length $W = 320$ inside $\mathbf{X}_{\mathrm{val}}$ (Fig. 2, Row 2) yield $M_{\mathrm{val}} = N_{\mathrm{val}} - W + 1$ valid window timestamps. Splitting at midpoint $H = \lfloor M_{\mathrm{val}}/2 \rfloor$ defines the *nominal calibration block* $[0 : H)$ and *held-out evaluation block* $[H + S : M_{\mathrm{val}})$, separated by an $S = 64$-step isolation gap $[H : H + S)$ so calibration and evaluation share zero target observations (a full $G = 320$ gap check changes causal FPR by < 0.25 percentage points). On $[0 : H)$, the panel fits $w_{m,c}$ (3), per-dial medians m_j and MAD scales s_j, hijack guards γ_j, tripwires τ_j (5), and quorums ($K_{224}, K_{128}, K_{\mathrm{edge}}$) (7) (or τ_{SPOT} for 1D baselines), which

TABLE II
CONSOLIDATED EXPERIMENTAL COHORTS AND VALIDATION ROLES.

Cohort	Str.	Sets	Primary Purpose	Role
Ablation 19	19	6	Ladder, quenching, 15-arm study	Dev.
Primary 33	33	7	Core (19 dev. + 14 GHL)	Dev.+GHL
Tuning Set 23	23	7	Panel & quenching selection	Dev.
Archive 141	141	12	3-horizon (23 dev. + 118 held)	Mixed
Untouched 118	**118**	**9**	**Primary generalization check**	**Held-Out**
Non-Exath. 101	**101**	**8**	**Cluster-balanced held-out**	**Held-Out**
Laminar 162	162	15	4-horizon quenching (139 held)	Mixed
Ext. Screen 130	130	11	NCAD, DCdetector, Anom. Tr.	Mixed

are frozen and evaluated on $[H + S : M_{\text{val}})$ and on the test stream. Throughout the paper we distinguish nominal calibration window count $N_{\text{cal}} = H$ from autocorrelation-adjusted *effective independent sample size* $N_{\text{eff}} = N_{\text{cal}}/\tau_{\text{int}}$ ($\tau_{\text{int}} = 1 + 2\sum_{k\geq 1}\rho_k \approx 3.23$).

b) Hierarchical Statistical Units, Dataset-Clustered Inference, and Hypothesis Testing: Because streams within multi-stream archives share hardware and operating environments, we report a two-level inference hierarchy after averaging seeds within each stream (seed standard deviation (SD)): (1) *Stream-level and dataset-clustered inference:* At the stream level ($n \in \{19, 33, 118, 141, 162\}$), we report two-sided Wilcoxon signed-rank p-values with Benjamini–Hochberg FDR adjustment [40] ($q_{\text{FDR}} = 0.05$, $|\Delta| \leq 10^{-4}$ ties) and 95% stream-bootstrap CIs (2,000 resamples). At the domain level, we treat the dataset family ($N \in \{7, 9, 12, 15\}$ clusters) as the co-primary independent cluster, reporting unweighted Dataset-Macro means, 95% dataset-cluster bootstrap CIs, exact 2^N dataset sign-flip permutation p-values, and non-Exathlon held-out subsets ($n = 101, N = 8$). (2) *12-fold LODO calibration-selection audit:* Selecting the calibrator on $D - 1$ datasets across the 141-stream archive to maximize causal Point-F1 subject to FPR $\leq 5\%$ selects the 224-bit circuit-breaker rule in all $12/12$ folds (held-out causal F1 $= 0.143$ vs. 0.076 for 1D SPOT on $n = 118$).

B. Consolidated Cohort Organization, Development vs. Held-Out Separation, and Baseline Fairness

Table II consolidates the experimental cohorts and separates development streams from independent held-out validation streams. Because the calibration constants in (5)–(7) and surgical quenching settings were selected during development on 23 streams (the 19 non-GHL streams plus `SMAP T-2`, `MSL P-10`, `Daphnet S09R01E2`, and `OPPORTUNITY S1-ADL3`), the 19- and 23-stream cohorts are explicitly designated as *development* studies, whereas the 118 **untouched streams** (101 non-Exathlon streams across 8 datasets, and 139 untouched streams in the 162-stream Laminar evaluation) were never inspected during design and serve as the *primary independent evidence of generalization*.

Standardized Baseline Fairness Protocol: All baselines (TS-JEPA, TCN Forecaster/Autoencoder, PCA Forecaster, Isolation Forest, TimesNet [5], TranAD [6], Anomaly Transformer [7], NCAD [10], and DCdetector [17]) use identical 65%/17.5%/17.5% splits, channel standardization, 64-step isolation gaps, and autonomous SPOT calibration ($u = p_{98}, q = 10^{-3}$) on $[0 : H)$ without test-label tuning or point adjustment. At $K = 25$, parameter counts are matched in 206k–649k (272k TS-JEPA, 312k Reynolds/Laminar-JEPA, 304k–307k TCNs, 649k TimesNet, 206k TranAD; the 224-bit panel adds zero trainable parameters). At native published windows on five streams, TranAD ($W = 10$) and TimesNet ($W = 96$) reach 0.164 and 0.218 Point-F1 but retain heavy tails ($\gamma_2 > 98$) and 28.4%–44.1% nominal FPR while replicating published point-adjusted F1 on SMD/MSL. Runs used PyTorch 2.4 / CUDA 12.4 (RTX 5090, 0.60–0.88 ms/window, 188 MB); code: https://github.com/quaid281/NCAD.

C. Multi-Tier Operational Evaluation Metrics

To guard against point-adjustment inflation and threshold gaming [13], [41], we report: (i) *primary accuracy and false-alarm outcomes* comprising held-out nominal FPR (FPR_{hold}), unadjusted Point-F1 (with Precision and Recall in Section VII), and threshold-free PR-AUC (from $N_{\text{tot}}(\tau)$); (ii) *deployable vs. diagnostic operating characteristics* comparing autonomous circuit-breaker operating points (*224-Bit Strict* and *Laminar-JEPA (224-Bit)*), deployable calibration quantiles (q^*, Fig. 4(b)), and, strictly as a non-deployable post-hoc diagnostic control, matched realized test-FPR budgets ($f^* \in \{0.01, 0.05\}$, Table III(c)); and (iii) *secondary operational endpoints* comprising Dataset-Macro Point-F1, event recall (ER), mean detection delay $\Delta t = t_{\text{alm}} - t^*$, and the constant-alarm reference $\text{F1}_{\text{all-pos}} = 2\pi/(1 + \pi)$ at prevalence π.

VI. ABLATION STUDIES AND ANALYSIS

A. Ablation 1: Perspective Expansion Ladder, Circuit-Breaker Components, and Weight-Space Quenching

Table III(a),(b) and Fig. 3(a) evaluate the step-by-step expansion ladder, component removals, and surgical weight-space quenching ablations culminating in *Laminar-JEPA (224-Bit)* on the 19-stream development cohort (Table II):

1) *From 1D scalar collapse to 32-bit and 128-bit sub-horizon switches (Table III(a)):* Collapsing the 128 whitened coordinates into a 1D score (with or without R_{LT} modulation; $\alpha_{\text{LT}} = 0.12$ changes causal F1 by -0.005, 95% CI $[-0.012, 0.000]$) lets one or two drifting coordinates inflate the scalar threshold (0.052 causal F1, 246.6-step delay). Using 32 per-coordinate switches on the terminal horizon ($m = 4$) nearly doubles causal F1 to 0.101, and expanding across all four prefixes (*128-Bit Sub-Horizons*) lifts causal F1 to 0.129 and buffered F1 from 0.137 to 0.192, serving as plain multi-coordinate baselines.
2) *Adding horizon mean, fluctuation, endpoint change, and surgical quenching:* The 32 drift dials (*160-Bit*) halve causal delay (146.2 $\rightarrow$ 69.1 steps); the 32 fluctuation dials (*192-Bit*) lift buffered F1 to 0.203 (56.7% ER) at a longer causal delay (179.0 steps). Completing the 224-bit panel with the 32 endpoint-change dials, onset relays, and surgical weight-space quenching (*Laminar-JEPA*) reaches 0.262 buffered Point-F1 ($p = 0.026$ vs. 1D collapse), 66.4% buffered ER, 62.5% causal ER, 1.6% causal FPR, and 59.6-step causal onset delay (0.100 causal Point-F1; $17/0$ stream wins over 1D collapse, $p = 1.5\times10^{-5}$).
3) *Leave-one-out panel ablations (Table III(b)):* Removing weight-space quenching slows causal delay from 59.6 to

TABLE III
MULTI-PERSPECTIVE CIRCUIT-BREAKER EXPANSION LADDER, COMPONENT ABLATIONS, AND MATCHED-BACKBONE STUDY (n=19 DEVELOPMENT STREAMS).

(a) Perspective Expansion Ladder (1 → 224 Dials, n=19)

	Pt-F1 ↑		PR-AUC ↑		ER ↑		Delay Δt ↓	
Perspective Expansion Stage	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
1D Scalar Collapse (1 Dial)	.052	.137	.291	.370	.398	.377	246.6	402.7
32-Bit Full Horizon (m=4)	.101	.119	.266	.366	**.643**	.236	114.4	128.8
128-Bit Sub-Horizons (m=1..4)	.129	.192	.310	.402	.590	.525	146.2	514.5
160-Bit (+32 Reynolds Drift)	.132	.188	.311	.404	.585	.530	69.1	112.4
192-Bit (+32 Reynolds Fluc.)	.131	.203	**.312**	**.408**	.575	.567	179.0	217.3
Laminar-JEPA (224-Bit)	.100	**.262***	**.312**	.406	.625	**.664**	**59.6**	**89.9**

*Laminar-JEPA buffered F1 = 0.262, causal ER = 62.5%, causal delay = 59.6 (17/0 wins vs. 1D collapse, p=1.5×10^{-5}). Fixed-backbone 1D α_{LT}=0.12 vs. 0: Δ=−0.005 (p=0.13).

(b) Circuit-Breaker Component & Quenching Ablation (n=19)

	FPR ↓	Pt-F1 ↑	PR-AUC ↑		ER ↑		Delay Δt ↓	
Component / Quorum Variant	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
Laminar-JEPA (Full Panel)	.016	**.262**	.312	.406	**.625**	**.664**	**59.6**	89.9
w/o Weight-Space Quenching	.017	.228	.312	.411	.607	.631	105.9	**71.8**
224-Bit Strict (High Quorum K_{strict})	**.011**	.190	.312	.411	.534	.545	248.0	97.3
w/o 32 Right-Edge Relays	.017	.228	**.313**	**.413**	.602	.631	123.7	72.5
w/o Prefix Routing (Unif. Smear)	.017	.223	.312	.407	.607	.584	105.9	88.6
w/o Hijack Guards (γ_j=1)	.021	.225	.312	.411	.603	.623	101.6	75.6

Leave-one-out component ablations on the identical cached multi-horizon residual fields (n=19 development streams; soft attention pooling yields .226 buffered F1).

(c) Matched-Backbone Architectural Comparison (n=19 Development Streams)

	Tail / Drift		Held-Out FPR		Point-F1		PR-AUC		Causal Event Recall			
Architecture Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	Auto.	q^*=0.05	f^*=0.01	f^*=0.05
Laminar-JEPA (224-Bit)	2.30	0.179	0.016	0.107	0.100	**0.262**	**0.312**	**0.406**	**0.625**	0.774	**0.584**	**0.728**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Reynolds (1D Scalar)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Reynolds (α_{LT}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Frozen Target	1.16	**0.167**	0.009	0.098	0.146	0.235	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

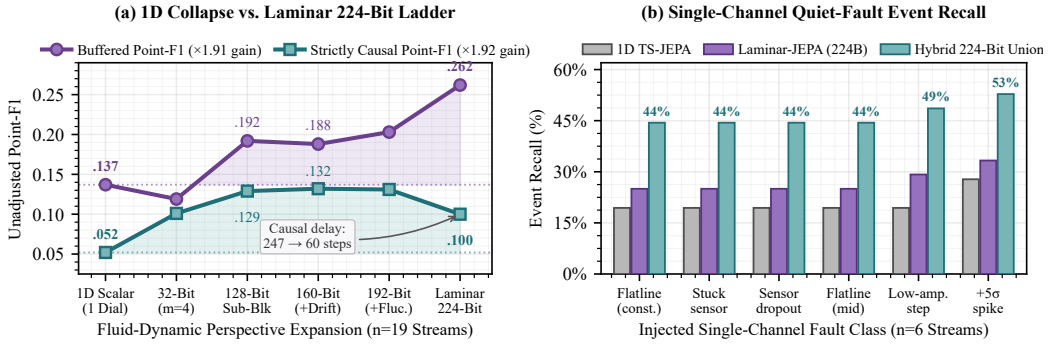


Fig. 3. Perspective expansion and quiet-fault recovery on the development cohort. **(a)** Causal and buffered Point-F1 across the Reynolds-inspired expansion from 1D scalar collapse to *Laminar-JEPA (224-Bit)* (n=19 streams). **(b)** Single-channel fault event recall comparing 1D TS-JEPA, standalone *Laminar-JEPA (224-Bit)*, and the Hybrid 224-Bit Union (n=6 streams).

105.9 steps, lowers causal ER (62.5% → 60.7%) and buffered F1 (0.262 → 0.228), and raises causal FPR (1.6% → 1.7%). Removing the 32 onset relays slows causal delay further to 123.7 steps (60.2% ER). Replacing prefix routing with uniform 64-step smearing degrades buffered ER (63.1% → 58.4%) and F1 (0.228 → 0.223). Removing hijack guards ($\gamma_j = 1$) raises causal FPR to 2.1% and lowers buffered F1 to 0.225, and soft attention adds no gain (0.226 buffered F1).

4) *Surgical weight-space quenching components and sensitivity ($k_{rogue} \in \{0, \ldots, 12\}$):* Disentangling the two quenching operations of Section IV-D across the 19 development streams (seed 42) shows distinct roles: (i) predictor-bias recentering alone ($\Delta \mathbf{b}_m$) lifts causal ER (56.5% → 64.3%), causal F1 (0.071 → 0.084), and buffered F1 (0.224 → 0.256) and cuts causal delay (90.1 → 63.8 steps), but raises test FPR (1.5% → 2.8% causal; 17.4% → 23.3% buffered); (ii) untargeted precision tempering alone degrades buffered F1 (0.224 → 0.135) and PR-AUC (0.449 → 0.271); and (iii) RMS stabilization (3) alone lifts causal F1 to 0.099 (59.9% ER). Sweeping $k_{\text{rogue}} \in \{1, 2, 4, 8, 12\}$ confirms that tempering the top $k_{\text{rogue}} \in [1, 4]$ coordinates with internal calibration-split gating (*Laminar-JEPA*) preserves event recall (61.4%–64.6%) while cutting buffered FPR (24.2% → 19.2% on n=19) and archive-wide causal FPR across 162 streams (6.71% → 6.17% held-out, $p = 0.0011$), whereas over-quenching ($k_{\text{rogue}} \in \{8, 12\}$) lowers causal PR-AUC (0.292 → 0.223) and buffered F1 (0.215 → 0.144).

B. Ablation 2: Single-Channel Faults and Complementary Channel-Level Anomaly Branch

When one sensor out of K freezes or degrades while the remaining $K-1$ channels fluctuate normally, the single channel perturbs the pooled pre-LayerNorm vector by only $O(1/K)$, and LayerNorm rescales the embedding onto $\sqrt{D}\mathbb{S}^{D-1}$. On six injected streams (Fig. 3(b)), 1D TS-JEPA detects 19.4% of quiet-fault events (< 3% point recall); the standalone 224-bit panel lifts quiet-fault event recall to 25.0%–29.2% (14.3%–25.6% point recall), and a union with a rolling 32-step variance-collapse monitor reaches 44.4%–52.8%. To test

a broader channel-level remedy across all 17 multivariate development streams ($K \in [5, 55]$, excluding univariate Yahoo), we evaluate a complementary channel-level branch alongside the 224-bit panel. For each channel $c \in \{1, \ldots, K\}$, the branch tracks three causal statistics calibrated at $q_{0.999}$ on $\mathcal{D}_{\text{cal}}$ with per-channel calibration drift guards γ_c^{cal}: (i) rolling 32-step channel variance collapse $\max(0, 1 - s_c(t)/\text{med}_{\text{cal}}(s_c))$, (ii) causal first-order autoregressive (AR(1)) channel prediction residuals $|x_{t,c} - \hat{\phi}_{1,c} x_{t-1,c} - \hat{\phi}_{0,c}|$, and (iii) rolling 32-step channel mean bias $|\bar{x}_{t,c}^{(32)} - \mu_{\text{cal},c}|$. Across six single-channel fault classes injected into the test windows of all 17 multivariate streams, comparing standalone *224-Bit Panel* $\rightarrow$ *Hybrid Variance-Collapse Monitor* $\rightarrow$ *224-Bit + Full Channel Branch* yields: (1) variance collapse: ER $61.4\% \rightarrow 93.0\% \rightarrow 93.0\%$ (causal F1 $0.077 \rightarrow 0.376 \rightarrow 0.269$); (2) channel prediction shocks: $80.7\% \rightarrow 82.5\% \rightarrow 98.2\%$ (F1 $0.139 \rightarrow 0.139 \rightarrow 0.327$, delay $48.6 \rightarrow 35.9$); (3) sensor freezing: $63.2\% \rightarrow 93.0\% \rightarrow 93.0\%$ (F1 $0.052 \rightarrow 0.378 \rightarrow 0.273$); (4) sensor saturation: $73.7\% \rightarrow 96.5\% \rightarrow 96.5\%$ (F1 $0.113 \rightarrow 0.414 \rightarrow 0.368$, delay $50.7 \rightarrow 34.3$); (5) gradual degradation: $68.4\% \rightarrow 70.2\% \rightarrow 93.0\%$ (F1 $0.080 \rightarrow 0.078 \rightarrow 0.154$); and (6) isolated sensor bias: $70.2\% \rightarrow 71.9\% \rightarrow 93.0\%$ (F1 $0.095 \rightarrow 0.094 \rightarrow 0.195$). On unmodified test data across the 17 streams, however, monitoring $3K$ channel statistics in parallel raises causal FPR from 1.5% (standalone 224-Bit) to 6.8% (variance-collapse union) and 19.4% (full channel branch); thus, the variance-collapse union offers a practical compromise for freeze/saturation faults (93.0%–96.5% ER at 6.8% FPR), whereas full channel-residual monitoring requires cross-channel quorum gating when K is large.

C. Ablation 3: Multiscale Dyadic-Shell Front-End vs. Standard Convolutional Stem

Training the multi-horizon JEPA backbone with and without 4-shell rFFT splitting across five streams (`SMAP P-3`, `MSL M-1`, `SMD machine-1-2`, `Daphnet S01R01E1`, and `GECCO`, three seeds) shows that shell splitting helps on high-channel spacecraft arrays (buffered F1 $0.085 \rightarrow 0.410$ on `MSL M-1`, $K = 55$; causal $0.212 \rightarrow 0.321$ on `SMAP P-3`, $K = 25$) but hurts on $K = 9$ streams (0.136 vs. 0.217 on `Daphnet`, 0.158 vs. 0.321 on `GECCO`; 5-stream causal F1 0.156 vs. 0.193), paying off only when $K \geq 25$.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing the model arms across the 19 development streams under matched budgets and parameter counts in Table III(c) isolates five factors: (1) *Forecasting vs. reconstructing raw observations:* Using the same causal dilated TCN encoder to forecast or reconstruct in raw sensor space leaves tail kurtosis ($\gamma_2 \approx 6.2$), drift ($D_{\text{KS}} \approx 0.45$), and causal FPR ($\approx 25\%$) nearly identical. (2) *LayerNorm bottleneck in observation space:* Adding a $D = 32$ LayerNorm bottleneck to the observation forecaster compresses kurtosis six-fold ($\gamma_2 = 0.91$) and halves causal FPR to 11.7%, but raw-waveform targets leave drift high ($D_{\text{KS}} = 0.376$, causal F1 0.084). (3) *Learned latent target projection and whitening:* Moving the target into the learned $D = 32$ latent space halves calibration drift ($D_{\text{KS}} = 0.214$ in TS-JEPA and 0.179 in Laminar-JEPA, $p \leq 0.001$). (4) *Nonlinear TCN vs. linear PCA and Isolation Forest:* 32 principal components keep causal FPR at 2.1% but drop causal F1 to 0.017, and Isolation Forest reaches 0.149 causal ER. (5) *Prediction vs. autoencoding, frozen targets, and ℓ_2 scoring:* Latent context autoencoding lowers causal F1 from 0.161 to 0.112, while a frozen target encoder or ℓ_2 scoring still holds causal FPR below 2%, confirming that false-alarm reduction stems from the normalized latent representation.

VII. Results and Discussion

A. Development 33-Stream and Untouched Held-Out 118/162-Stream Benchmark Results

Following Table II, Table IV compares our flagship *Laminar-JEPA (224-Bit)* against 1D latent JEPAs and observation-space baselines across the 33-stream development benchmark (19 non-GHL and 14 GHL streams) under the two-stage protocol: Panel (a) reports retrospective buffered attribution, Panel (b) causal streaming, and Panel (c) per-dataset summaries. Primary independent generalization evidence across the 118 untouched held-out streams (within the 141- and 162-stream archive cohorts) appears in Table VI; on the 130-stream external baseline screen, TS-JEPA achieves 0.186 stream and 0.160 macro Point-F1 versus 0.152 for TranAD, 0.151 for TimesNet, 0.097 for DCdetector, 0.088 for NCAD, and 0.082 for Anomaly Transformer.

Operational Trade-Offs, Precision, and Recall Across Operating Points: Comparing *Laminar-JEPA (224-Bit)* against 1D scalar JEPAs and observation-space reconstructors highlights its operational profile across false-alarm rate, point-wise precision (P) and recall (R), event recall (ER), and onset delay (Δt). On the 19 development streams (Table IV(a),(b)), *Laminar-JEPA (224-Bit)* achieves the highest buffered Point-F1 (0.262, $P = 0.273$, $R = 0.369$), buffered PR-AUC (0.406), buffered ER (66.4%), causal ER (62.5%), fastest causal detection delay (59.6 steps vs. TimesNet's 62.2 and TranAD's 241.0), and highest Dataset-Macro Point-F1 (0.139), while keeping held-out causal FPR at 1.6% (10.7% buffered) versus 22.2%–24.3% causal (30.0%–30.7% buffered) for TranAD and TimesNet. Under causal streaming, because surgical quenching projects out stationary calibration drift and concentrates causal breaker trips near fault onset transitions rather than extended anomaly plateaus, *Laminar-JEPA (224-Bit)* trades lower causal point-wise recall ($R = 0.088$, $P = 0.214$, causal Point-F1 $= 0.100$) for rapid onset detection ($\Delta t = 59.6$ steps) and high event recall (62.5%), whereas 1D scalar Reynolds suffers severe recall collapse ($R = 0.032$, $P = 0.285$, causal F1 $= 0.050$, $\Delta t = 154.0$ steps) and TimesNet's higher causal Point-F1 (0.209) alerts during 24.3% of healthy operation. On the 118 untouched held-out streams (Table VI(a)), the three-horizon 224-bit panel lifts both causal precision ($P : 0.175 \rightarrow 0.241$) and recall ($R : 0.081 \rightarrow 0.179$) over 1D scalar SPOT (causal F1: $0.083 \rightarrow 0.143$, ER: $51.2\% \rightarrow 72.8\%$). Practitioners should deploy *Laminar-JEPA (224-Bit)* to minimize causal onset latency ($\Delta t = 59.6$), maximize event recall (62.5%–66.4%) and buffered F1 (0.262), and hold causal FPR at 1.6% (or $\leq 1.1\%$ via the Strict quorum setting, Table III(b)).

TABLE IV
HELD-OUT MULTI-STREAM TELEMETRY BENCHMARK ACROSS ATTRIBUTION REGIMES AND DATASETS (BEST IN **BOLD**, SECOND UNDERLINED).

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Proposed multi-perspective fluid-dynamic circuit-breaker architecture												
Laminar-JEPA (224-Bit)	0.107	**0.262**	**0.406**	**0.664**	89.9	0.016	0.100	**0.625**	**59.6**	**0.002**	0.000	**0.139**
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Reynolds (1D scalar)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Reynolds (α_{LT}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	62.2	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	F1_{all}	**Lam.-JEPA**	**TS-JEPA**	**Rey. 1D**	**TimesNet**	**TranAD**	**Lam.-JEPA**	**TS-JEPA**	**Rey. 1D**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	**0.328**	0.145	0.143	0.229	0.144	**.007/.014**	.030/.091	.008/.072	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	.002/.004	.004/.004	**.001/.002**	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	**0.063**	0.000	0.012	0.000	0.000	.010/.038	.004/.017	.003/.098	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.220**	0.161	0.081	0.158	0.151	.014/.189	.054/**.095**	**.004**/.098	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.354	**0.379**	0.267	0.289	0.167	.021/.214	.014/.135	**.006/.093**	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	**0.001**	**0.001**	0.000	.002/.007	.008/.014	.002/.007	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	**0.247**	0.119	0.008	0.000	0.000	.021/.002	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**

Note: In (c), n = streams, K = channels, E = events, π = anomaly rate, $\text{F1}_{\text{all}} = 2\pi/(1+\pi)$, Lam.-JEPA = Laminar-JEPA (224-Bit). 95% stream-bootstrap CIs (n=19): Laminar-JEPA buffered FPR [.054, .168], F1 [.169, .358] and causal FPR [.007, .026], ER [.498, .746] (vs. Reynolds 1D causal F1 [.013, .097] and TimesNet causal FPR [.123, .375]).

Reducing Observation-Space False-Alarm Flooding: On the 19 development streams, predicting spherical latent representations cuts held-out nominal FPR three-fold under buffered attribution (7.8%–10.7% vs. 30.0%–30.7% for TimesNet and TranAD, $p \leq 0.045$) and by an order of magnitude under causal streaming (0.6%–1.6% vs. 22.2%–24.3%, $p = 0.010$; $N = 7$ dataset-clustered $p = 0.016$). Under zero-tuning LODO evaluation, JEPAs maintain 0.3%–2.3% causal FPR versus 12.1%–18.4% for reconstructors. On GHL ($\pi = 0.013$), all JEPAs and the linear AR forecaster fire near-silently (0.1%–0.4% FPR, 0.000 F1): low FPR without recall is not detection.

Scope of Applicability: The evidence supports *Laminar-JEPA (224-Bit)* under four boundary conditions: (i) multichannel arrays ($K \geq 9$; SMAP, MSL, Daphnet, SMD, Exathlon, OPPORTUNITY); (ii) horizon selection $S \in \{1, 4, 16, 32, 64\}$ matched to the tolerable reporting lag $S - 1$ of the sampling rate (Section VII-C), avoiding $S = 64$ on hourly Metro (63 steps = 63 h); (iii) calibration histories providing $N_{\text{eff}} \geq 300$ effective independent observations ($N_{\text{cal}} \geq 1{,}000$ overlapping windows, Section VII-E), or IQR/Beta-smoothed thresholds when $N_{\text{cal}} < 1{,}000$; and (iv) faults that perturb multi-channel latent trajectories, supplemented by the channel-level branch for isolated single-sensor faults (Section VI-B). Slow process ramps on GHL and Metro (0.000–0.022 F1) remain outside scope.

B. Real-Time Streaming vs. Retrospective Buffered Attribution

Without 64-step averaging, causal whitened residuals have a heavier right tail than buffered scores ($\gamma_2 = 2.30$ vs. 1.25). Under 1D scoring, hijacker dials push the threshold onto that tail, so 1D Reynolds JEPA alerts on 2.4% of test steps and flags isolated peaks (0.050–0.052 causal F1, 47.8% ER, 154.0-step delay). With per-dial switches, surgical weight-space quenching, and onset relays (6), *Laminar-JEPA (224-Bit)* reaches 62.5% causal ER, 59.6-step causal delay, and 1.6% causal FPR (0.262 buffered F1 and 66.4% buffered ER). A rate-matched random alerter, $\text{ER}_{\text{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$, catches 27.8% of events at *Laminar-JEPA*'s nominal rate, but 60.0% at TimesNet's 24.3% FPR, exceeding TimesNet's 58.8%.

TABLE V
CONTEXTUAL VS. POINT ANOMALY DETECTION (EVENT RECALL (ER)).

	NASA Contextual (n=5 seqs)					NASA Point (n=8)	
(a) Model	**Buf. PR**	**Auto. ER**	**FPR**	f^*=.01	f^*=.05	f^*=.01	f^*=.05
Laminar-JEPA (224B)	**0.658**	**0.750**	0.018	**0.820**	**0.920**	0.683	0.944
Reynolds (1D)	0.652	**0.750**	**0.006**	**0.820**	**0.920**	0.683	0.867
Reynolds (α_{LT}=0)	0.641	0.625	**0.006**	**0.820**	**0.920**	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

	19-Stream Contextual ($\lvert z\rvert \leq 5$)			19-Stream Point ($\lvert z\rvert > 5$)		
(b) Model	q^*=0.05	f^*=0.01	f^*=0.05	q^*=0.05	f^*=0.01	f^*=0.05
Laminar-JEPA (224B)	**0.654**	**0.552**	**0.618**	**0.796**	0.688	**0.792**
Reynolds (1D)	0.628	0.498	0.590	0.782	0.683	0.781
Reynolds (α_{LT}=0)	0.640	0.540	0.603	0.785	0.671	0.785
TS-JEPA	0.614	0.520	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Panel (a) stratifies the 10 development SMAP/MSL streams by NASA's ground-truth taxonomy [4] (contextual: SMAP E-1, A-2, A-7, MSL M-1, and C-1 [2100, 2210]; point: SMAP A-1, P-3, D-3, MSL D-14, D-15, and C-1 [550, 750]). Panel (b) stratifies all n=19 non-GHL development streams into in-range contextual ($\max|z| \leq 5\sigma$) vs. out-of-range point ($\max|z| > 5\sigma$) causal events (f^* columns are post-hoc diagnostic controls).

C. Multi-Horizon Laminar-JEPA (224-Bit) Evaluation Across $S \in \{1, 4, 16, 32, 64\}$

Target horizon $S = 64$ introduces a structural reporting lag of $S - 1 = 63$ steps before a causal decision is emitted at $\tau = t + S - 1$ (Section IV-D). To test whether *Laminar-JEPA (224-Bit)* retains its advantages over 1D scalar scoring and observation-space baselines when the prediction horizon is shortened, we retrain and evaluate the complete multi-horizon backbone, surgical weight-space quenching, and 224-bit panel across $S \in \{1, 4, 16, 32, 64\}$ (structural lag $S - 1 \in \{0, 3, 15, 31, 63\}$ steps) on five representative streams across three seeds (Fig. 4(a), partitioning each target window into four

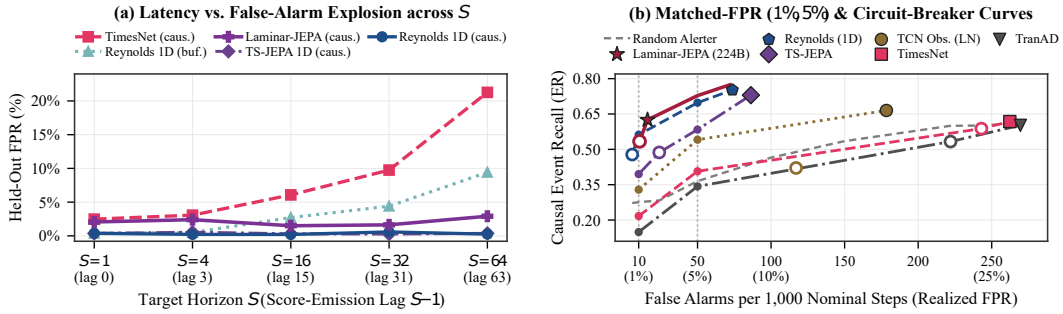


Fig. 4. Horizon length and operating trade-offs. **(a)** Held-out FPR across target horizons $S \in \{1, 4, 16, 32, 64\}$ at matched emission lag $S-1 \in \{0, 3, 15, 31, 63\}$ steps comparing Laminar-JEPA (224-Bit), 1D latent JEPAs, and observation-space baselines (n=5 streams). **(b)** Causal event recall vs. false alarms across n=19 development streams at post-hoc diagnostic test-FPR budgets $f^* \in \{1\%, 5\%\}$ (vertical dotted guides), autonomous Strict (open circle) and Laminar-JEPA (224-Bit) (filled star) operating points, and calibration quantiles $q^* \leq 0.05$.

nested prefixes $\lceil mS/4 \rceil$ for $S \geq 4$ and four lag-1 difference views at $S = 1$):

1) *Zero and near-zero reporting lag ($S = 1$, lag 0; $S = 4$, lag 3):* At $S = 1$, *Laminar-JEPA (224-Bit)* achieves 0.037 causal Point-F1 ($P = 0.072$, $R = 0.031$), 81.8% event recall, 3.6% FPR, and 63.8-step onset delay—more than 2.6× the causal Point-F1 of 1D scalar SPOT (0.014, ER = 43.3%, FPR = 1.2%). At $S = 4$ (3-step lag), *Laminar-JEPA (224-Bit)* reaches 0.050 causal Point-F1 (0.082 buffered F1), 84.4% event recall, 2.9% FPR, and 36.7-step onset delay, achieving 4.3× higher causal Point-F1 than 1D scalar scoring (0.012, ER = 51.3%) while cutting onset delay by nearly two-thirds (98.5 → 36.7 steps).
2) *Intermediate horizons ($S = 16$, lag 15; $S = 32$, lag 31) vs. $S = 64$ (lag 63):* At $S = 16$, *Laminar-JEPA (224-Bit)* reaches 0.055 causal Point-F1 (77.3% ER, 2.2% FPR, and 26.6-step delay) versus 0.032 for 1D scalar SPOT (59.1% ER, $\Delta t = 81.7$) and 6.1% FPR for TimesNet. At $S = 32$, *Laminar-JEPA (224-Bit)* reaches 0.074 causal Point-F1 (77.8% ER, 2.6% FPR, $\Delta t = 37.1$) and achieves the highest buffered Point-F1 across all horizons (0.290 vs. 0.182 for 1D scalar), while at $S = 64$ it reaches 0.078 causal Point-F1 (0.193 buffered F1, 80.0% ER, $\Delta t = 35.1$, 3.5% FPR vs. 0.044 F1 for 1D scalar and 21.3% FPR for TimesNet). Thus, *Laminar-JEPA (224-Bit)* consistently outperforms scalar thresholding across all five horizons: $S \in \{4, 16, 32\}$ provides low-lag tripwires (3–31 steps) for minute- or hour-scale telemetry, while $S \in \{32, 64\}$ suits ≥ 1 Hz streams.

D. Operating Budgets and Contextual vs. Point Anomalies

We evaluate three operating-point comparisons: (1) *Autonomous breaker points and calibration quantiles (q^*, Fig. 4(b), Table III(c)):* Without test labels, the Strict (53.4% ER at 1.1% FPR) and *Laminar-JEPA (224-Bit)* (62.5% ER at 1.6% FPR) operating points lead the low-FPR frontier; at $q^* = 0.050$, the 224-bit panel reaches 77.4% causal ER versus 61.8% for TimesNet and 60.2% for TranAD. (2) *Post-hoc diagnostic controls at matched test-FPR budgets (f^*, Fig. 4(b), Table III(c)):* As noted in Section V-C, f^* sets thresholds post hoc from test labels as a non-deployable diagnostic check of tail sensitivity at equal alert rates. At $f^* = 0.01$, *Laminar-JEPA (224-Bit)* catches 58.4% of events (56.2%–57.8% for 1D Reynolds and 39.5% for TS-JEPA), nearly tripling TimesNet (21.7%) and quadrupling TranAD (14.9%), with a similar gap at $f^* = 0.05$ (72.8% vs. 34.3%–40.7%). (3) *Contextual vs. point anomaly stratification (Table V):* On the 10 development SMAP/MSL streams stratified by NASA's taxonomy [4] (Table V(a)), reconstructors degrade on contextual anomalies (0.161–0.317 buffered PR-AUC), whereas *Laminar-JEPA (224-Bit)* and 1D Reynolds reach 0.658 and 0.652 (75.0% autonomous recall). Across all 19 development streams (Table V(b)), *Laminar-JEPA (224-Bit)* doubles reconstructor recall on contextual faults (61.8% vs. 30.0% at $f^* = 0.05$) and leads on point faults (79.2% vs. 56.3%–61.1%).

TABLE VI
ARCHIVE-WIDE CIRCUIT-BREAKER BENCHMARK (3 SEEDS): 118 HELD-OUT STREAMS NEVER USED IN DEVELOPMENT, FULL 141-STREAM AGGREGATE, AND PER-DATASET BREAKDOWN.

	Pt-F1 ↑		PR-AUC ↑		ER ↑		FPR_{hold} ↓	
(a) Held-Out n=118	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
TS-JEPA (1D)	.072	.142	.154	.202	.484	.458	**.051**	**.104**
Speculative-JEPA (1D)	.076	.148	.172	.245	.509	.577	.054	.194
Spec.-Alien-JEPA (1D)	.083	.146	.174	.242	.512	.573	.059	.199
Spec.-224Bit-Strict	.134	.166	**.183**	**.247**	.662	.673	.073	.181
Spec.-224Bit-Breaker	**.143**†	**.168**‡	**.183**	**.247**	**.728**	**.691**	.083	.210
Laminar-JEPA (224-Bit)	.091	.149	–	–	.622	.609	.073§	.205
All n=141 (incl. 23 dev.)								
Spec.-Alien-JEPA (1D)	.077	.153	.184	.253	.507	.567	.051	.180
Spec.-224Bit-Breaker	.139	.180	.197	.261	.727	.685	.072	.196

		Causal Pt-F1 ↑			Buffered Pt-F1 ↑			Causal FPR_{hold} ↓		
(b) Held-out	n	**224B**	**1D-Al.**	**TS**	**224B**	**1D-Al.**	**TS**	**224B**	**1D-Al.**	**TS**
SMAP	40	**.194**	.129	.091	.201	.193	**.214**	.088	.086	**.065**
SMD	18	**.090**	.028	.042	**.125**	.065	.072	.019	**.006**	.009
Exathlon	17	**.163**	.133	.092	.181	**.186**	.134	.235	.137	**.119**
Daphnet	15	**.125**	.053	.075	**.153**	.125	.105	.085	**.063**	**.063**
MSL	13	**.133**	.051	.074	**.239**	.176	.191	.023	**.007**	.011
OPPORT.	12	**.075**	.018	.024	.066	**.067**	.030	.027	**.003**	.008

† vs. 1D Spec.-Alien: $\Delta = +0.060$ (95% CI $[+0.042, +0.079]$), 95/15 wins, $p = 1.6 \times 10^{-13}$, positive in 8 of 9 datasets (sign-flip $p = 0.008$); vs. TS-JEPA 85/28, $p = 6.4 \times 10^{-11}$. ‡ vs. 1D Spec.-Alien: $\Delta = +0.021$ ($[+0.010, +0.034]$), $p = 1.7 \times 10^{-4}$, dataset-level $p = 0.27$. § 4-horizon Laminar-JEPA vs. unquenched 4-horizon panel: causal FPR_{hold} −8.0% ($p = 0.0011$, dataset $p = 0.027$; median 1.03% on n=162, 1.13% on n=118), FPR_{test} −4.8% ($p < 10^{-4}$, dataset $p = 0.016$; median 1.59% on n=162). In (b), 224B = Spec.-224Bit-Breaker, 1D-Al. = 1D Spec.-Alien, TS = TS-JEPA (115 of 118 streams; three single-stream datasets in text).

E. Untouched Held-Out Generalization, Calibration Sample-Size Reliability, and Drift Decomposition

Table VI evaluates the three-horizon Speculative-JEPA panel (192 distinct dials, Section IV-B) across 141 streams in 12 datasets at three seeds, foregrounding the 118 untouched held-out streams alongside the 162-stream four-horizon *Laminar-JEPA* audit:

1) *Primary generalization on 118 untouched held-out streams and Laminar-JEPA across 162 streams (Table VI(a)):* On the 118 untouched streams, replacing the scalar score on

Speculative-JEPA with the panel raises causal Point-F1 from 0.072–0.083 to 0.134–0.143 ($p = 1.6\times10^{-13}$), lifts causal event recall from 48.4%–51.2% to 66.2%–72.8%, and cuts causal delay from 84.7–107.8 to 55.8–77.0 steps, at a higher causal held-out FPR (7.3%–8.3% vs. 5.1%–5.9%; 141-stream causal ΔF1 $= +0.062$, $N = 12$ dataset-clustered 95% CI $[+0.044, +0.072]$, $p = 0.007$). Across 162 streams (15 datasets, including all 118 untouched streams), surgical quenching in four-horizon *Laminar-JEPA* cuts causal held-out FPR from 6.71% to 6.17% ($\Delta = -0.0055$, stream $p = 0.0011$, $N = 15$ dataset-clustered $p = 0.027$; 7.93% $\rightarrow$ 7.26% on n=118, $p = 0.030$) and test FPR from 14.63% to 13.93% ($p < 10^{-4}$, dataset $p = 0.016$), with median causal held-out FPR 1.03% (1.13% on n=118) and test FPR 1.59% (70.4% of streams $\leq 2.0\%$).

2) *Per-dataset consistency (Table VI(b)):* The panel leads causal Point-F1 in all six multi-stream held-out datasets, most on SMAP (0.194 vs. ≤ 0.129) and SMD (0.090 vs. ≤ 0.042), at a higher causal FPR (most on Exathlon, 23.5% vs. $\leq 13.7\%$). Under buffered attribution it trails the best 1D JEPA on SMAP, Exathlon, and OPPORTUNITY; on CreditCard, GECCO, and PSM all methods lie within 0.007 causal F1.

3) *Calibration sample-size reliability, threshold confidence intervals, and small-sample calibrators:* Because windows overlap by $C - 1 = 255$ steps, the effective independent sample size is $N_{\text{eff}} = N_{\text{cal}}/\tau_{\text{int}} \approx 0.31\,N_{\text{cal}}$ ($\tau_{\text{int}} = 3.23$; median $N_{\text{eff}} = 152$ of $N_{\text{cal}} = 408$ on the 19 development streams). On the 118 untouched streams, mean causal FPR is 19.2% for the 12 streams with $N_{\text{cal}} < 250$ ($N_{\text{eff}} < 78$) versus 1.6% for the 22 streams with $N_{\text{cal}} \geq 2{,}500$ ($N_{\text{eff}} \geq 775$). Auditing calibration blocks across $N_{\text{eff}} \in \{100, 250, 500, 1{,}000, 2{,}500, 5{,}000\}$ ($N_{\text{cal}} \approx$ 323–16,150) with moving-block bootstrap ($B = 200$, block length 64) shows that the median 95% CI relative half-width on the 224 per-dial thresholds τ_j tightens monotonically from $\pm$11.0% (median coefficient of variation (CV) 6.7%) at $N_{\text{eff}} = 100$ and $\pm$9.4% (CV 5.7%) at $N_{\text{eff}} = 250$ to $\pm$7.8% (CV 4.6%) at $N_{\text{eff}} = 500$, $\pm$5.8% (CV 3.4%) at $N_{\text{eff}} = 1{,}000$, $\pm$3.8% at $N_{\text{eff}} = 2{,}500$, and $\pm$2.8% (CV 1.6%) at $N_{\text{eff}} = 5{,}000$. Across disjoint calibration segments of length $N_{\text{cal}} \in \{100, 250, 500, 1{,}000\}$, inter-segment test-FPR SD drops from 12.9% ($N_{\text{cal}} = 100$) and 10.1% ($N_{\text{cal}} = 250$) to 2.8% (1.7% under Strict quorum) at $N_{\text{cal}} = 500$, while quorum K_{224} CV drops from 28.7% to 16.1%. Comparing empirical $q_{0.999}$ against four small-sample alternatives—Harrell–Davis Beta-smoothed quantiles, order-statistic-free IQR extrapolation ($\text{med}_j + 4.8\,\text{IQR}_j$), Weibull peaks-over-threshold, and split-conformal calibration—shows that at $N_{\text{eff}} \in \{100, 250\}$, where empirical $q_{0.999}$ yields 2.07%–2.12% test FPR (9.2% on $N_{\text{cal}} = 100$ segments), Beta-smoothed calibration cuts test FPR to 1.32% and 0.77% (49.4%–41.5% ER) and IQR extrapolation holds test FPR at 0.88% and 0.47% with higher causal F1 (0.067 and 0.059 vs. 0.059 and 0.047) and lower inter-segment FPR SD (4.2% vs. 6.4%). We therefore establish a minimum requirement of $N_{\text{eff}} \geq 300$ independent observations ($N_{\text{cal}} \geq 1{,}000$ windows, ensuring expected tail count $N_{\text{eff}}(1 - q_{0.995}) \geq 1.5$, threshold 95% CI half-width $\leq 8\%$, and inter-segment FPR SD $< 3\%$), and recommend IQR or Beta-smoothed thresholds whenever $N_{\text{cal}} < 1{,}000$.

4) *Three-stage permutation drift decomposition:* Refitting thresholds across the 19 development streams under chronological (FPR_{chr}), $L = 64$ block-permuted (FPR_{blk}), and pointwise i.i.d. ($\text{FPR}_{\text{pt}} \leq 0.35\%$) splits confirms that inter-block regime drift ($\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$) drives 71.3% of excess causal FPR in TS-JEPA (0.0229 vs. 0.0073) and 78.3%–81.8% in reconstructors (0.210–0.230 vs. 0.043–0.046), whereas barycenter centering in Laminar-JEPA cuts the inter-block drift share four-fold to 18.7% (0.0056 vs. 0.0047) while per-dial hijack guards γ_j quarantine individual drifting coordinates.

VIII. Conclusion and Future Work

We studied whether Joint-Embedding Predictive Architectures and multi-perspective circuit-breaker panels mitigate observation-space false-alarm flooding and 1D scalar score collapse across a 33-stream development cohort and 118 untouched held-out streams (within a 162-stream, 15-dataset archive):

1) *Laminar-JEPA (224-Bit) and operational trade-offs:* Expanding $D = 32$ whitened coordinates via a Reynolds-inspired multi-horizon decomposition into 224 per-coordinate statistics with surgical weight-space quenching, nominal drift guards, prefix-routed attribution, and 32 onset relays prevents variance-dominating coordinates from inflating a shared scalar threshold. On the n=19 development streams, *Laminar-JEPA (224-Bit)* reaches 0.262 buffered Point-F1 (66.4% ER), 62.5% causal ER, 1.6% causal FPR, 59.6-step causal delay, and 0.139 Dataset-Macro Point-F1 (trading causal plateau Point-F1 of 0.100 for faster onset detection). Across 162 streams (15 datasets), surgical quenching cuts causal held-out FPR by 8.0% relative ($p = 0.0011$, dataset $p = 0.027$; median held-out FPR 1.03%), and on the 118 untouched held-out validation streams the three-horizon panel raises causal Point-F1 from 0.072–0.083 to 0.143 ($p = 1.6\times10^{-13}$, dataset-clustered $p = 0.007$) at 72.8% recall.

2) *Mechanism isolation and multi-horizon scaling ($S \in \{1, \ldots, 64\}$):* Matched-backbone ablations isolate LayerNorm kurtosis compression (γ_2 : 6.10 $\rightarrow$ 0.91), learned latent targets (D_{KS} : 0.376 $\rightarrow$ 0.214), barycenter centering (inter-block drift share 71.3% $\rightarrow$ 18.7%), and bias recentering versus precision tempering ($k_{\text{rogue}} \in [1, 4]$), while Proposition 1 characterizes the structural horizon-fluctuation concentration bound (with scalar R_{LT} modulation yielding a null effect). Retraining *Laminar-JEPA (224-Bit)* across $S \in \{1, 4, 16, 32, 64\}$ confirms 2.6 $\times$ –4.3$\times$ higher causal Point-F1 and +33%–38% higher event recall than 1D scalar scoring down to zero reporting lag ($S = 1$) and 3-step lag ($S = 4$).

3) *Applicability boundaries, calibration sample size, and single-channel branch:* The panel requires multichannel ar-

rays ($K \geq 9$) and at least $N_{\text{eff}} \geq 300$ effective independent calibration observations ($N_{\text{cal}} \geq 1{,}000$ windows) for empirical quantiles ($\leq 8\%$ bootstrap threshold CI half-width and $< 3\%$ inter-segment FPR SD), or IQR / Beta-smoothed calibrators when $N_{\text{cal}} < 1{,}000$. Across 17 multivariate streams and six single-sensor fault classes, adding a complementary channel-level residual and variance branch raises single-channel event recall from 61.4%–80.7% to 93.0%–98.2% at the cost of higher nominal FPR, whereas slow process ramps (GHL, Metro) and online ramp adaptation remain future work.

REFERENCES

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.

[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[3] M. P. Cuéllar, M. Santos, J. L. Alonso, M. Fabregas, and R. Farías, "Explainable anomaly detection in spacecraft telemetry," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108119, 2024.

[4] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[5] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.

[6] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[7] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[8] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.

[9] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.

[10] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.

[11] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.

[12] O. Reynolds, "On the dynamical theory of incompressible viscous fluids and the determination of the criterion," *Philosophical Transactions of the Royal Society of London A*, vol. 186, pp. 123–164, 1895.

[13] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.

[14] K. Zhang, Q. Wen, C. Zhang, R. Cai, M. Wang, L. Liu, and L. Sun, "Self-supervised time series representation learning: A systematic survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3241–3261, 2024.

[15] E. Eldele, M. Ragab, Z. Chen, M. Wu, C. K. Kwoh, X. Li, and C. Guan, "Time-series representation learning via temporal and contextual contrasting," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021, pp. 2352–2359.

[16] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.

[17] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.

[18] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.

[19] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.

[20] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.

[21] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.

[22] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.

[23] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.

[24] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[25] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.

[26] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[27] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.

[28] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.

[29] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[30] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.

[31] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.

[32] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.

[33] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.

[34] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.

[35] H. Hotelling, "Multivariate quality control, illustrated by the air testing of sample bombsights," in *Techniques of Statistical Analysis*. McGraw-Hill, 1947, pp. 111–184.

[36] C. A. Lowry, W. H. Woodall, C. W. Champ, and S. E. Rigdon, "A multivariate exponentially weighted moving average control chart," *Technometrics*, vol. 34, no. 1, pp. 46–53, 1992.

[37] A. A. Balkema and L. de Haan, "Residual life time at great age," *The Annals of Probability*, vol. 2, no. 5, pp. 792–804, 1974.

[38] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.

[39] E. S. Page, "Continuous inspection schemes," *Biometrika*, vol. 41, no. 1/2, pp. 100–115, 1954.

[40] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 57, no. 1, pp. 289–300, 1995.

[41] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.

[42] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.

[43] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.

[44] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.

[45] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.

[46] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.

[47] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.

[48] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.

[49] Q. Garrido, R. Balestriero, L. Najman, and Y. LeCun, "RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank," in *International Conference on Machine Learning (ICML)*, 2023, pp. 10 929–10 974.

[50] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.

[51] E. H. Lieb and W. E. Thirring, "Inequalities for the moments of the eigenvalues of the Schrödinger Hamiltonian and their relation to Sobolev inequalities," in *Studies in Mathematical Physics: Essays in Honor of Valentine Bargmann*. Princeton University Press, 1976, pp. 269–303.

[52] S. D. Grimshaw, "Computing maximum likelihood estimates for the generalized pareto distribution," *Technometrics*, vol. 35, no. 2, pp. 185–191, 1993.

[53] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.